

THE BURKHOLDER MARTINGALE TRANSFORM INEQUALITY IN LEAN

1. OVERVIEW

This note describes the formal Lean theorem

`Lp_Burkholder_inequality_martingaleTransform`

proved in `MartingaleTransforms.lean`. The result is the real-valued martingale transform inequality

$$\|(w \star_m f)_n\|_{L^p(\mu)} \leq (p^* - 1) \|f_n\|_{L^p(\mu)}$$

for every time n , where $1 < p < \infty$, f is a martingale, and w is a predictable multiplier bounded by 1 in absolute value.

In Lean the theorem is stated with the extended nonnegative exponent $p : \mathbb{R}_{\geq 0, \infty}$. The hypotheses include $1 < p$, $p \neq \infty$, and the proof uses `p.toReal` for the real exponent in the Burkholder functions.

2. BASIC DEFINITIONS

2.1. Martingale differences. For a discrete real-valued process $f : \mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$, the martingale difference sequence is defined by

$$d_0 f = f_0, \quad d_{n+1} f = f_{n+1} - f_n.$$

In Lean this is

$$\text{martingaleDiff } f \ 0 = f \ 0, \quad \text{martingaleDiff } f \ (n+1) = f \ (n+1) - f \ n.$$

2.2. Martingale transforms. Given a multiplier process $v : \mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$, the discrete martingale transform of f by v is

$$(v \star_m f)_n = \sum_{i=0}^n v_i d_i f.$$

The Lean definition is

$$\text{martingaleTransform } v \ f := \text{fun } n \Rightarrow (\text{Finset.range } (n + 1)).\text{sum fun } i \Rightarrow v \ i * \text{martingaleDiff } f \ i$$

The notation used in the file is

$$v \star_m f.$$

The increment identity proved in Lean is

$$(v \star_m f)_{n+1} - (v \star_m f)_n = v_{n+1}(f_{n+1} - f_n).$$

2.3. Predictability. The multiplier w is assumed to be strongly predictable with respect to a filtration $\mathcal{F}$:

$$\text{IsStronglyPredictable } \mathcal{F} \ w.$$

This is the formal version of the usual condition that w_{n+1} is $\mathcal{F}_n$ -measurable. Predictability is what allows the proof to pull the multiplier out of conditional expectations when showing that the linear terms have expectation zero.

2.4. **The auxiliary transforms X and Y .** The proof uses two transforms associated to w :

$$w_n^+ = w_n + 1, \quad w_n^- = w_n - 1.$$

In Lean these are `plusOne w` and `minusOne w`. The notation

$$X_n[w, f] = ((w + 1) \star_m f)_n, \quad Y_n[w, f] = ((w - 1) \star_m f)_n$$

is introduced by scoped notation. These satisfy the algebraic identities

$$\frac{X_n + Y_n}{2} = (w \star_m f)_n, \quad \frac{X_n - Y_n}{2} = f_n.$$

The final theorem uses these identities together with the definition of the Burkholder function v .

3. THE MAIN MARTINGALE TRANSFORM THEOREM

The Lean theorem has the following mathematical content.

Theorem 1 (Burkholder martingale transform inequality). *Let $(\Omega, \mathcal{A}, \mu)$ be a finite measure space and let $(\mathcal{F}_n)_{n \in \mathbb{N}}$ be a filtration. Let $1 < p < \infty$. Suppose $f = (f_n)$ is a real-valued martingale, $f_n \in L^p(\mu)$ for every n , and $w = (w_n)$ is strongly predictable with*

$$|w_n(\omega)| \leq 1$$

for every n , almost everywhere. Then for every n ,

$$\|(w \star_m f)_n\|_{L^p(\mu)} \leq (p^* - 1) \|f_n\|_{L^p(\mu)},$$

where

$$q = \frac{p}{p-1}, \quad p^* = \max\{p, q\}.$$

In Lean, the constant is written as

$$\text{ENNReal.ofReal } (\text{Burkholder.pStar } p.\text{toReal} - 1).$$

4. THE BURKHOLDER FUNCTION v

The proof is organized around the Burkholder expression

$$v(p, x, y) = \left| \frac{x + y}{2} \right|^p - |p^* - 1|^p \left| \frac{x - y}{2} \right|^p.$$

This is the Lean definition `Majorants.v`. Substituting

$$x = X_n[w, f], \quad y = Y_n[w, f]$$

and using the identities above gives

$$v(p, X_n, Y_n) = |(w \star_m f)_n|^p - |p^* - 1|^p |f_n|^p.$$

Thus the inequality

$$\int v(p, X_n, Y_n) d\mu \leq 0$$

is exactly the p -power form of the martingale transform estimate. The last step of the Lean proof unfolds the definition of v , converts the integral inequality into an inequality of real L^p -norms, and then converts back to `eLpNorm`.

5. THE MAJORANT THEOREM `EXISTS_MAJORANT_P_G_1`

The analytic heart of the formalization is

`exists_majorant_p_g_1.`

For every real $p > 1$, it produces functions

$$u, \quad u_x, \quad u_y : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$$

and a constant $C \geq 0$ such that:

- u, u_x, u_y are continuous on all of $\mathbb{R}^2$;
- u has p -growth:

$$|u(x, y)| \leq C(|x|^p + |y|^p);$$

- the selected derivatives have $(p - 1)$ -growth:

$$|u_x(x, y)|, |u_y(x, y)| \leq C(|x|^{p-1} + |y|^{p-1});$$

- u satisfies the tangent/concavity inequality in martingale-transform directions:

$$hk \leq 0 \implies u(x + h, y + k) \leq u(x, y) + u_x(x, y)h + u_y(x, y)k;$$

- u majorizes v :

$$v(p, x, y) \leq u(x, y);$$

- u is nonpositive on the region $xy \leq 0$:

$$xy \leq 0 \implies u(x, y) \leq 0.$$

5.1. Why the majorant u is important. The role of the majorant u is to turn a norm inequality into a supermartingale-type inequality. The function v directly contains the desired estimate, but by itself it does not have the concavity property needed to use the martingale structure. The majorant u supplies this missing property:

- (1) Since $v \leq u$, it is enough to prove that the expectation of $u(X_n, Y_n)$ is nonpositive.
- (2) The tangent inequality for u shows that

$$u(X_{n+1}, Y_{n+1}) \leq u(X_n, Y_n) + u_x(X_n, Y_n)(X_{n+1} - X_n) + u_y(X_n, Y_n)(Y_{n+1} - Y_n).$$

- (3) The condition $|w| \leq 1$ implies

$$(X_{n+1} - X_n)(Y_{n+1} - Y_n) \leq 0,$$

which is exactly the hypothesis needed for the tangent inequality.

- (4) By predictability of w and the martingale property of f , the linear terms have integral zero.
- (5) Therefore,

$$\int u(X_{n+1}, Y_{n+1}) d\mu \leq \int u(X_n, Y_n) d\mu.$$

- (6) At the initial time, $X_0 Y_0 \leq 0$, so $u(X_0, Y_0) \leq 0$. Hence $\int u(X_n, Y_n) d\mu \leq 0$ for every n .

Combining this with $v \leq u$ gives

$$\int v(p, X_n, Y_n) d\mu \leq 0,$$

and the definition of v then yields the desired martingale transform inequality.